

Diagnosing the Evaluation Distribution: Argument-Taxonomy Distillation and an English Pivot for Climate Stance Detection in Hindi and Bengali

Koushik Deb^{1,2}

¹Indian Institute of Information Technology Kalyani, Kalyani, West Bengal, 741235, India

²Accenture, Kolkata, West Bengal, India

Abstract

We describe our submission to a shared task on detecting the stance of Hindi and Bengali YouTube comments towards the claim “*climate change and global warming is a serious concern*”, given only English training resources. Our central finding is procedural rather than architectural: the decisive step was measuring the evaluation data before choosing a transfer strategy. Inspection of the released files shows that they consist of *machine-translated English* comments—no romanised code-mixing, no emoji, three mentions of India across 500 Bengali comments, and recurring arguments about a science communicator’s credentials and the “97 % consensus” figure. Building for that distribution instead of for organic Indic social media reversed our design. We formalise a 27-node climate-argument taxonomy that extends the CARDS scheme with an India-specific branch and an explicit NONE inventory, and use it as a generative schema, as auxiliary supervision, and as an annotation scaffold. We distil a nine-model large-language-model panel over the unlabelled evaluation set into compact encoders. Finally we exploit the forensic finding a second time by back-translating the evaluation set into English and classifying with an English-only encoder, which sidesteps the cross-lingual problem rather than solving it; this is worth +0.10 macro-F1 over the best multilingual configuration, verified by a paired test over 300 shared data splits. Our submitted system attains macro-F1 0.922 (Hindi) and 0.915 (Bengali) on a held-out development set. We state clearly that those development labels come from a language-model judge panel rather than human annotation, and we report both the reliability evidence and its limits. We also document four negative results, including a methodological trap in which comparing ensemble strategies by unpaired means overstated one apparent gain and reversed the sign of another.

Keywords

stance detection, low-resource languages, Hindi, Bengali, knowledge distillation, argument mining, model calibration

1. Introduction

Stance detection asks how a text positions itself with respect to a fixed claim, which is distinct from sentiment: a furious comment and an approving one can take the same stance, and a polite one can take none at all. Given a YouTube comment in Hindi or Bengali, the task is to assign FAVOUR, AGAINST, or NONE with respect to the claim “*climate change and global warming is a serious concern*”. The organisers supply English-only training resources and evaluate on Hindi and Bengali, so cross-lingual transfer is the nominal challenge. Runs are ranked by macro-F1, and more than eighty runs were submitted to the track.

Two properties of the task shaped our approach. First, macro-F1 weights every class equally, so the AGAINST class—about 15 % of the permitted English pool—carries a full third of the score. A system that never confidently predicts AGAINST is arithmetically capped near 0.45 however well it performs elsewhere. Second, and less obviously, the evaluation data is not what the task description leads one to expect. We therefore begin with a diagnostic rather than a model.

Contributions.

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EMAIL: koushikdeb2009@gmail.com (K. Deb)



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1. A forensic characterisation of the evaluation set (Section 4) establishing that it consists of machine-translated English comments, together with the four design decisions that follow.
2. A 27-node climate-argument taxonomy (Section 5) extending CARDS [1] with an India-specific contrarian branch, a mirrored pro-climate branch, and an explicit NONE inventory, used in three distinct roles.
3. An English-pivot classifier (Section 6.4) worth +0.10 macro-F1 over the best multilingual system.
4. Transductive panel distillation and macro-F1-aware calibration, each measured separately (Sections 6.2 and 6.5).
5. Four negative and methodological results (Section 9) that we believe are more useful to report than a further architecture.

2. Related Work

Stance detection was established as a benchmark task by SemEval-2016 Task 6 [2], and extended to unseen targets by zero-shot formulations [3]. For climate discourse specifically, GWSD provides English news sentences with crowd stance annotations [4], and the CARDS scheme organises contrarian claims into a hierarchy of super-claims and sub-claims [1], which we adopt and extend.

Cross-lingual transfer for Indic languages typically relies on multilingual encoders: XLM-R [5], MuRIL [6], pre-trained on Indian-language corpora including transliterated pairs, and IndicBERTv2 [7]. An alternative reformulates classification as entailment against a hypothesis [8], using XNLI-trained models in multilingual settings [9]; we include this as a baseline. Surveys document rapid adoption of large language models for stance detection [10], generally as direct predictors or as annotators.

We instead use language models as a *teacher*, following knowledge distillation [11]: panel labels on unlabelled target data are distilled into the compact encoder we actually submit, which keeps every run a locally reproducible classifier rather than a live API call. Our projection channel adapts a hybrid framework for multilingual zero-shot classification [12] aligning sentence embeddings with knowledge-graph label embeddings; where that work uses ConceptNet [13] we substitute our task-specific taxonomy, and we retain its Kolmogorov–Arnold [14] alignment.

3. Task and Data

The evaluation data comprises two files of 500 comments each, for Hindi and Bengali, with columns ID and COMMENT and identifiers 1–500 in both files. Runs are submitted as one file per language carrying up to three classifier columns.

The permitted training resources are two English stance corpora: **GWSD** [4], 2 300 news sentences each annotated by eight crowd workers, and **SemEval-2016 Task 6** [2] restricted to the target “*Climate Change is a Real Concern*” (395 tweets). Other public data, machine translation and synthetic generation are explicitly permitted, and the organisers encourage making the training set more appropriate for the task.

Merging and de-duplicating yields 2 442 rows (NONE 1 092, FAVOUR 983, AGAINST 367). For GWSD we retain the full crowd vote distribution rather than only the majority label: an 8–0 vote and a 5–3 vote are different training signals, and the soft target regularises the small AGAINST class at no cost. Mean GWSD annotator agreement is 0.72. Only 15.0 % of the pool is AGAINST, and within the SemEval subset just 3.8 % (15 of 395). Under macro-F1 this is not a detail but the substance of the task.

4. Characterising the Evaluation Set

Before training any model we measured the released files. Table 1 summarises the result. Organic Indian YouTube comments are short, emoji-dense and substantially Roman-script; these comments are none of those things.

Table 1

Measurements over all 1 000 evaluation comments.

Signal	Hindi	Bengali
Comments naming one science communicator	63	96
Roman-script code-mixing	0	0
Comments containing any emoji	0	0
Comments mentioning India	2	3
Median length (characters)	146	147

The register is uniformly formal and the arguments are drawn from a United States-centric repertoire: attacks on a presenter’s credentials (“*not a scientist, a mechanical engineer*”), disputes over the “97 %” consensus figure, channel-merchandise spam, creationist framing (“*why did God kill the dinosaurs*”), and comments indicating a school assignment (“*who else is here for online classes*”). Numerals appear in Devanagari and Bengali digits and English names are transliterated. We conclude that the evaluation set consists of English YouTube comments machine-translated into Hindi and Bengali.

Matching the multiset of numerals per comment—normalised across Devanagari, Bengali and ASCII digits—shows that the two language files are different comment samples rather than translations of the same 500 English comments, so no cross-file alignment shortcut exists.

Four consequences. This reverses the natural design. First, translate-test becomes the primary channel rather than a fallback: back-translation recovers near-original English, the native domain of GWSD, of SemEval, and of every strong English encoder (Section 6.4). Second, one sentence of provenance context in an annotation prompt resolves whole families of otherwise ambiguous items. Third, synthetic data should imitate translationese—pure native script, formal register, no emoji, transliterated names—rather than authentic Indian slang. Fourth, the argument inventory to cover is the source community’s, not the target language’s; our India-specific taxonomy branch, designed before this measurement, accordingly became a minor slice, and we report it as an ablation rather than as the centrepiece.

5. Argument Taxonomy

Cross-lingual stance transfer for climate discourse fails less because of language mismatch than because of *argument-inventory* mismatch. GWSD and SemEval encode a United States contrarian idiom; translating them yields fluent Hindi sentences about Al Gore, which is grammatically correct and pedagogically useless. We therefore pivot through the argument rather than through the language.

Our taxonomy contains 27 nodes—13 AGAINST, 7 FAVOUR, 7 NONE—with 228 cue phrases across English, Hindi and Bengali. The **CARDS branch** (6 nodes) follows Coan et al. [1]: not happening; not human-caused; impacts not harmful; solutions will not work; science unreliable; and one node we added, *attacks on the messenger*, which the measurements in Section 4 showed to be among the most frequent arguments actually present. The **India-specific contrarian branch** (7 nodes) covers climate colonialism and whataboutism, development-first arguments, local-weather naturalism, cyclical or religious cosmology, media conspiracy, elite hypocrisy, and population-rather-than-climate. The **pro-climate branch** (7 nodes) is grounded in lived experience: floods and air quality, agricultural harm, health, intergenerational duty, demands for action, support for mitigation, and trust in science. The **NONE inventory** (7 nodes) covers channel praise, pure questions, neutral factual statements, unrelated politics, spam, ambivalence, and—again added after the measurements—comments indicating a class assignment.

Two annotation traps are encoded explicitly, because they are where systems and annotators most often diverge. *Sentiment is not stance*: “*Shame on this government, they have destroyed our rivers!*” is FAVOUR despite maximally negative sentiment. *Praise of the video is not agreement with the claim*: “*very*

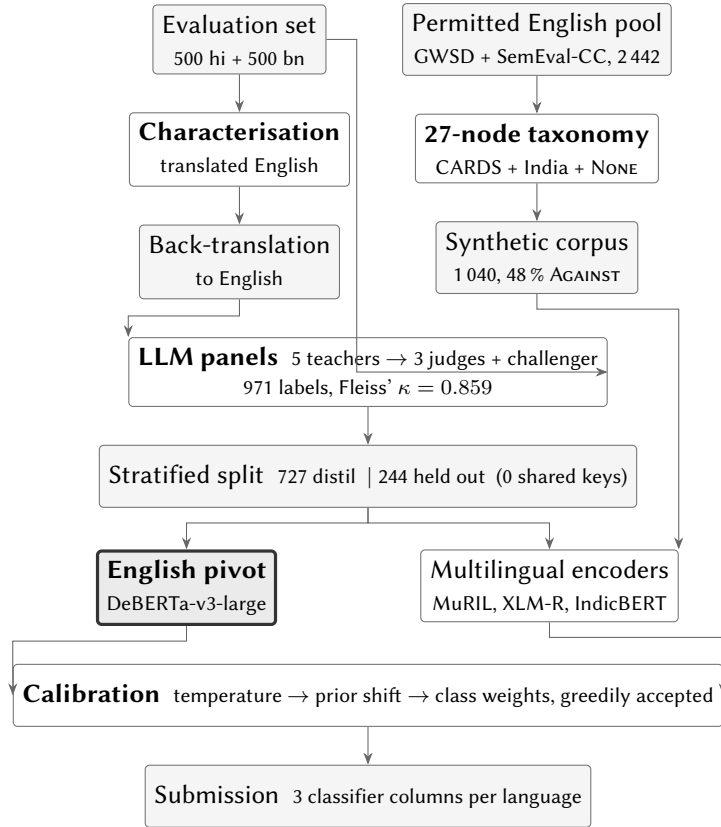


Figure 1: The pipeline. Characterising the evaluation set (upper left) is what makes the English pivot available; everything downstream of it is a consequence of that measurement rather than a prior design choice.

informative, thank you” is NONE even under a climate-alarm video.

The taxonomy is used in three roles, and that triple use is what makes it a contribution rather than a lexicon appendix: as a generative schema (Section 6.1), as auxiliary supervision (Section 6.3), and as the reasoning scaffold in the annotation prompt (Section 6.2).

6. Method

Figure 1 shows the full pipeline.

6.1. Taxonomy-conditioned synthetic augmentation

We walk the product node $\times$ language $\times$ script $\times$ register and prompt a language model for realistic comments in each cell. Because the stance is fixed by the cell, the corpus is class-balanced by construction. Following Section 4 we generate in translationese style. The result is 1 040 comments covering all 27 nodes, of which 48 % are AGAINST. Provenance columns (node, branch, language, register, generator) are retained for ablation, and near-duplicates are removed by token Jaccard similarity within each (node, language) cell.

Figure 2 shows why this matters, and contains an unplanned validation. The permitted English pool is 15 % AGAINST; the synthetic corpus was deliberately built at 48 %; and the evaluation set, as later labelled by our judge panel, turned out to be 44 % AGAINST. The synthetic distribution was therefore constructed to match a target we had not yet measured, and landed within four points of it.

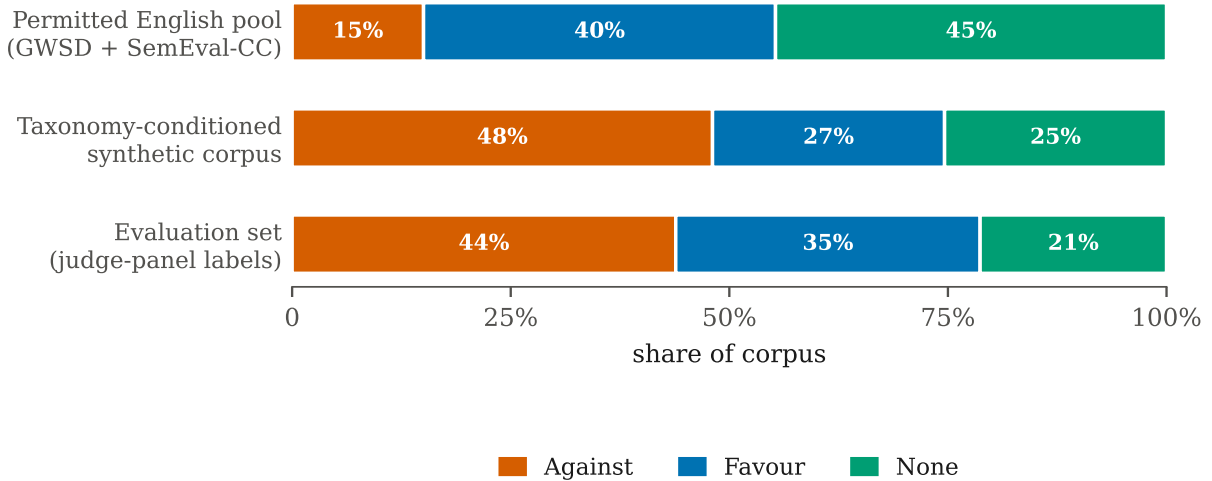


Figure 2: Class composition of the three corpora. The permitted training pool is dominated by NONE and FAVOUR; the evaluation set is AGAINST-heavy. Under macro-F1 this mismatch, not the language gap, is what caps a naive system.

Table 2

Panel reliability. Teacher-panel κ is reported per language.

Measure	Value
Teacher panel Fleiss’ κ (5 models)	0.767 / 0.749
Judge panel Fleiss’ κ (3 models)	0.859
Judge labels accepted (of 1 000)	971
Adversarial overturn rate	2.8 %
Teacher–judge agreement	92.7 %
Teacher macro-F1 against judge labels	0.917

6.2. Transductive panel distillation

The entire *unlabelled* evaluation set is available at inference time. Using it is transduction, not leakage: no organiser-provided label is read at any point. We run two disjoint panels over all 1 000 comments.

A **teacher panel** of five inexpensive models from five different laboratories labels every comment. Each model must name a taxonomy node *before* emitting a stance, which is chain-of-thought constrained to a fixed and auditable vocabulary. A **judge panel** of three stronger models then adjudicates, reading the Indic original together with its English back-translation, with a fourth model adversarially challenging every proposed label. Panels are disjoint by construction and the code refuses to run if any model identifier appears in both roles.

The 971 accepted labels are split, stratified by language and label, into **727 distillation rows** and a **244-row held-out development set**, with an assertion that the two share no (language, identifier) key. Every number in Section 8 is measured on that held-out set.

6.3. Argument-aware encoder

Encoders carry two heads over a shared representation: a three-way stance head trained with class-weighted cross-entropy plus a soft-label KL term, and an auxiliary 28-way argument-node head with weight $\lambda = 0.3$. The auxiliary signal is free—synthetic rows are node-labelled by construction and the panels predict a node for every real comment—and it pushes the shared representation towards argument structure rather than surface lexical cues, which is the axis along which we need it to generalise. Rows without a node are masked out of the auxiliary loss. We use class weighting rather than oversampling, because duplicating rows distorts soft targets whereas reweighting does not.

6.4. The English pivot

This is the component that mattered most, and it follows directly from Section 4. We back-translate all 1 000 evaluation comments into English and fine-tune DeBERTa-v3-large [15] on an English-only pool consisting of the 2 442 GWSD and SemEval rows together with the back-translated judge-labelled rows. Rather than asking a multilingual encoder to bridge English training data and Indic evaluation data, we move the evaluation data to where the training data already lives.

6.5. Macro-F1-aware calibration

Taking $\arg \max_y p(y \mid x)$ maximises accuracy, not macro-F1. We apply, in order, temperature scaling [16], prior-shift correction by the Saerens–Latinne–Decaestecker expectation-maximisation procedure [17], and a coordinate search over per-class multiplicative weights. Each stage is *accepted greedily*, retained only if it does not lower development macro-F1. This is not a formality: on Bengali, prior correction would have taken one channel from 0.781 to 0.200 and was rejected automatically. We additionally floor and damp the expectation-maximisation update, because on an under-confident model the textbook procedure drives $p(\text{AGAINST})$ to zero and removes the class entirely.

7. Experimental Setup

We evaluate MuRIL base and large [6], XLM-R base and large [5], IndicBERTv2 [7], DeBERTa-v3-large [15], and LaBSE [18] as a frozen encoder. Maximum sequence length is 128 and the random seed is fixed throughout. Base models were trained on 32 CPU cores; large models on a single rented RTX 4090 in bfloat16, at a total compute cost of USD 1.11. All submitted runs are local checkpoints requiring no API call at inference.

We also evaluate a **projection channel** adapting the hybrid framework of Deb et al. [12]. LaBSE embeddings are projected into a semantic space built from the taxonomy’s gloss-and-cue embeddings by a hybrid of a Kolmogorov–Arnold network [14] and support-vector regression, and stance is read from cosine similarity aggregated over nodes. Here the taxonomy replaces ConceptNet [13] as the knowledge source, which is both domain-matched and considerably richer than a bare three-way label space. Finally we include a training-free entailment baseline [8] using a multilingual XNLI model [9].

8. Results

Table 3 and Figure 3 give the main result. The English pivot dominates every multilingual configuration by a wide margin. Paired over 300 shared half-splits against the best pre-pivot system, the calibrated pivot gains +0.099 on Hindi (50.2 standard errors, winning 100 % of splits) and +0.115 on Bengali (58.9 standard errors, 100 %).

Before adopting a gain of that size we verified that it was not leakage. The English distillation pool shares zero (language, identifier) keys with the development set, no development comment text appears verbatim in any training text, and the GWSD and SemEval pool contains no evaluation comments at all.

Component contributions. Calibration is worth +0.019 to +0.023 on the pivot channel and +0.04 to +0.08 on base encoders. Model scale is worth +0.05 to +0.08: MuRIL-base to MuRIL-large, and XLM-R-base at 0.717 to XLM-R-large at 0.836. The auxiliary node head and the soft-label term are both retained in the submitted configuration.

Submitted runs. Both language files carry the calibrated English pivot as the primary classifier, the English-pool DeBERTa variant as the second, and XLM-R-large on native script as the third. All three are trained classifiers. The three columns agree on 87.2 % of Hindi comments and 86.4 % of Bengali comments.

Table 3

Macro-F1 on the 244-row held-out development set (123 Hindi, 121 Bengali), which shares no row with any training data. Submitted runs in bold. Per-class F1 is given for the submitted primary run.

System	hi	bn
<i>English pivot</i>		
DeBERTa-v3-L + calibration	0.922	0.915
without calibration	0.904	0.892
DeBERTa-v3-L, English-only pool	0.886	0.877
<i>Native-script encoders</i>		
XLM-R-large	0.836	0.812
MuRIL-large	0.807	0.790
IndicBERTv2	0.804	0.791
MuRIL-base, distilled	0.758	0.781
XLM-R-base, distilled	0.717	0.707
<i>Other channels</i>		
Argument-KG projection	0.763	0.722
Claim-conditioned entailment	0.438	0.404
Best pre-pivot configuration	0.810	0.779
<i>Per-class F1, submitted primary run</i>		
FAVOUR	0.935	0.921
AGAINST	0.963	0.961
NONE	0.870	0.863

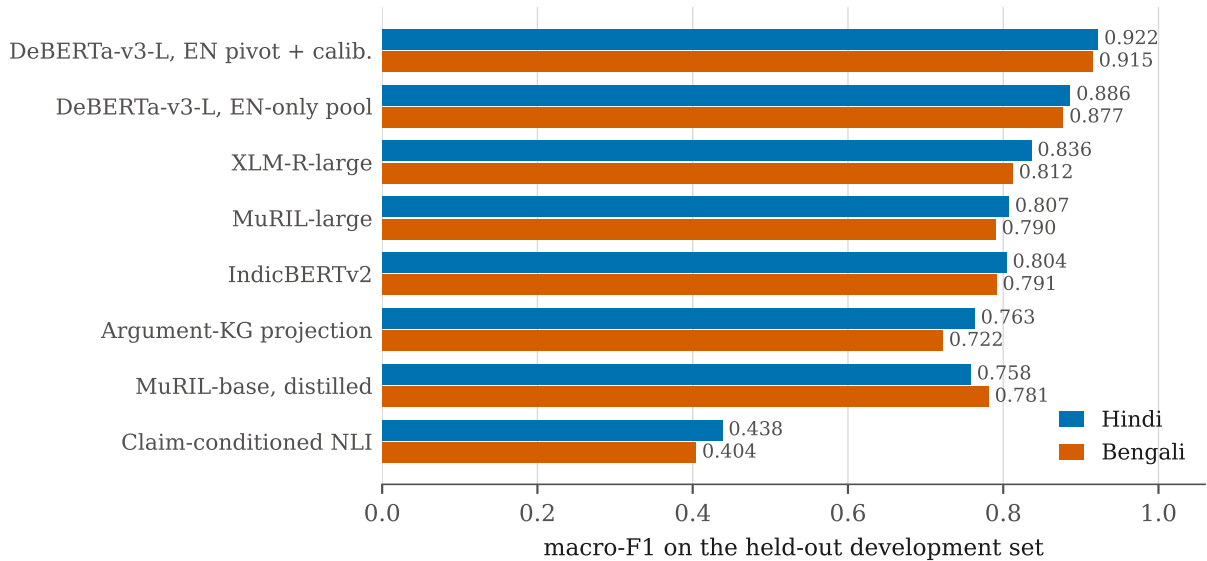


Figure 3: Held-out development macro-F1 by system. The two highest bars are English-pivot models; every native-script encoder, including three large ones, falls below them.

9. Negative and Methodological Results

More in-distribution labels hurt. Refitting the distilled encoder on all 971 judge labels, rather than the 727 that exclude the development set, made it *worse*: 0.655 on Hindi and 0.684 on Bengali against 0.758 and 0.781, despite then training on the very rows it was scored against. The likely cause is checkpoint selection. With an internal early-stopping split drawn from a pool that is 92 % synthetic and translated text, the selected epoch is chosen against the wrong distribution. More data is not automatically better when it changes what one early-stops on.

Every ensemble lost to the best single model. Measured against the calibrated pivot, all tested blends were worse, from -0.009 to -0.035 , with most winning 0 % of 300 splits. With roughly 120 development rows per language, combination weights cannot be estimated reliably, and a single strong model is the safer choice. We report this because ensembling is the default expectation in shared-task submissions.

Unpaired strategy comparison is misleading. Our strategy selector initially ranked configurations by mean held-out macro-F1 over independently drawn half-splits. Re-evaluated on identical splits, one apparent Hindi gain of $+0.015$ shrank to $+0.005$, and an apparent Bengali gain of $+0.023$ *reversed sign* to -0.002 . At this data size any ensembling decision must be made on paired differences over shared splits.

An evaluation-harness bug nearly cost us the best system. Our trainer’s in-training development score for the English pivot read 0.574, and a retry read 0.331, so we briefly abandoned the idea. Both figures were artefacts: the harness evaluated using a text column that held the Indic original, so an English-only model was being scored on Devanagari input. The prediction path had been correct throughout. We report this because the failure mode—a silent column mismatch producing a plausible-looking low score rather than an error—is easy to reproduce and hard to notice.

10. Error Analysis

The submitted system’s weakest class is NONE in both languages (0.870 and 0.863, against ≥ 0.96 for AGAINST). This matches where the taxonomy predicted difficulty. NONE is not a semantic category but the absence of a stance, and it collects channel praise, questions, neutral facts and ambivalence—categories whose boundary with FAVOUR rests on a codebook decision rather than on the text. The most frequent confusions are NONE predicted as FAVOUR, where polite engagement is read as agreement, and AGAINST predicted as FAVOUR, where sarcasm survives translation poorly.

The most frequent arguments assigned by the panels were demands for action, science-unreliable, attacks on the messenger, and channel praise. That attacks on the messenger ranks fourth in Bengali is a direct vindication of adding that node after the measurements of Section 4.

11. Limitations

Development labels are not human gold. This is the most important caveat. Our 244 development labels come from the judge panel, so every figure in Table 3 measures agreement with that panel rather than with ground truth. The panel’s internal agreement is high ($\kappa = 0.859$) and a human reader reviewed and endorsed a fifteen-comment sample, but that review was not blind and therefore yields no independent agreement statistic. If the panel’s reading of the NONE boundary differs systematically from the organisers’, our official score will be lower than reported here, and our calibration—fitted to the panel’s class priors—may transfer imperfectly. We regard a blind human annotation of 50–150 comments as the single highest-value extension of this work.

The pivot requires machine translation at inference. The primary run consumes back-translated English. The classifier itself is a local checkpoint with no API dependency, but reproducing the run requires either the back-translated files, which we release, or a re-translation step. Machine translation is a standard and organiser-encouraged step for this task, but it is a pipeline dependency and we state it plainly.

Synthetic data reflects model priors. The 1 040 generated comments encode a language model’s *model* of Indian climate discourse, which is not the same thing as that discourse.

Small development set. Roughly 120 rows per language. All ensembling conclusions are correspondingly weak, which is itself one of our findings.

12. Conclusion

The lesson we would carry to another cross-lingual shared task is procedural: measure the evaluation distribution before choosing a transfer strategy. Here the nominal problem was Hindi and Bengali social media, the actual problem was translated English, and recognising that was worth more than every architectural choice we made combined. The argument taxonomy, the synthetic corpus and the panel labels are released so that the components can be reused and, where we were wrong, corrected.

Declaration on Generative AI

The author declares the following uses of generative artificial intelligence in this work.

In the research pipeline, generative AI is a studied component rather than an incidental aid, and its role is described in full in Sections 6.1–6.2. Specifically: large language models generated the 1 040-comment synthetic training corpus; a five-model teacher panel and a three-model judge panel produced the pseudo-labels distilled into the submitted classifiers and the development labels used for calibration; and language-model-based machine translation produced the Hindi and Bengali training translations and the English back-translations on which the primary submitted run operates. No organiser-provided gold label was used at any point. The limitations of relying on model-generated development labels are stated in Section 11.

In preparing this manuscript, the author used an AI assistant for code implementation, experiment orchestration, and drafting and editing of the text. All experimental design decisions, the forensic analysis, and the interpretation of results are the author’s own; the author verified all reported numbers and takes full responsibility for the content of the publication.

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